EXP NO:4,DATE:11-2-2024

IMPLEMENT K-NEAREST NEIGHBOUR ALGORITHM TO CLASSIFY THE DATA SET. PRINT BOTH CORRECT AND WRONG PREDICTIONS

Aim

To classify the data set using k-Nearest Neighbour algorithm and to print correct and wrong predictions

Components Required

Jupyter Notebook (Anaconda 3) – Software PC

In [4]:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.neighbors import KNeighborsClassifier
import matplotlib.pyplot as plt
from mlxtend.plotting import plot_decision_regions
df = pd.read_csv("bmi.csv")
df.head()
```

Out[4]:

	Height	Weight	Index
0	174	96	1
1	189	87	0
2	185	110	1
3	195	104	1
4	149	61	1

In [7]:

```
X = df[["Height", "Weight"]]
y = df["Index"]
# Split the data
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_s

# Define and train the model
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)

# Predict and evaluate
y_pred = knn.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.9
```

In [13]:

```
for i in range(len(X_test)):
 print("Test data:", X_test.iloc[i], "| True label:", y_test.iloc[i], "|")
if y_test.iloc[i] == y_pred[i]:
 print("Correct prediction")
else: print("Wrong prediction")
```

```
Test data: Height    191
Weight     79
Name: 14, dtype: int64 | True label: 0 |
Test data: Height    153
Weight    107
Name: 15, dtype: int64 | True label: 1 |
Test data: Height    190
Weight     95
Name: 26, dtype: int64 | True label: 0 |
Test data: Height    171
Weight    152
Name: 38, dtype: int64 | True label: 1 |
Test data: Height    181
Weight    111
Name: 47, dtype: int64 | True label: 1 |
Test data: Height    179
Weight    152
Name: 29, dtype: int64 | True label: 1 |
Test data: Height    185
Weight    110
Name: 2, dtype: int64 | True label: 1 |
Test data: Height    178
Weight     52
Name: 31, dtype: int64 | True label: 1 |
Test data: Height    157
Weight     56
Name: 42, dtype: int64 | True label: 0 |
Test data: Height    195
Weight    104
Name: 3, dtype: int64 | True label: 1 |
Correct prediction
```

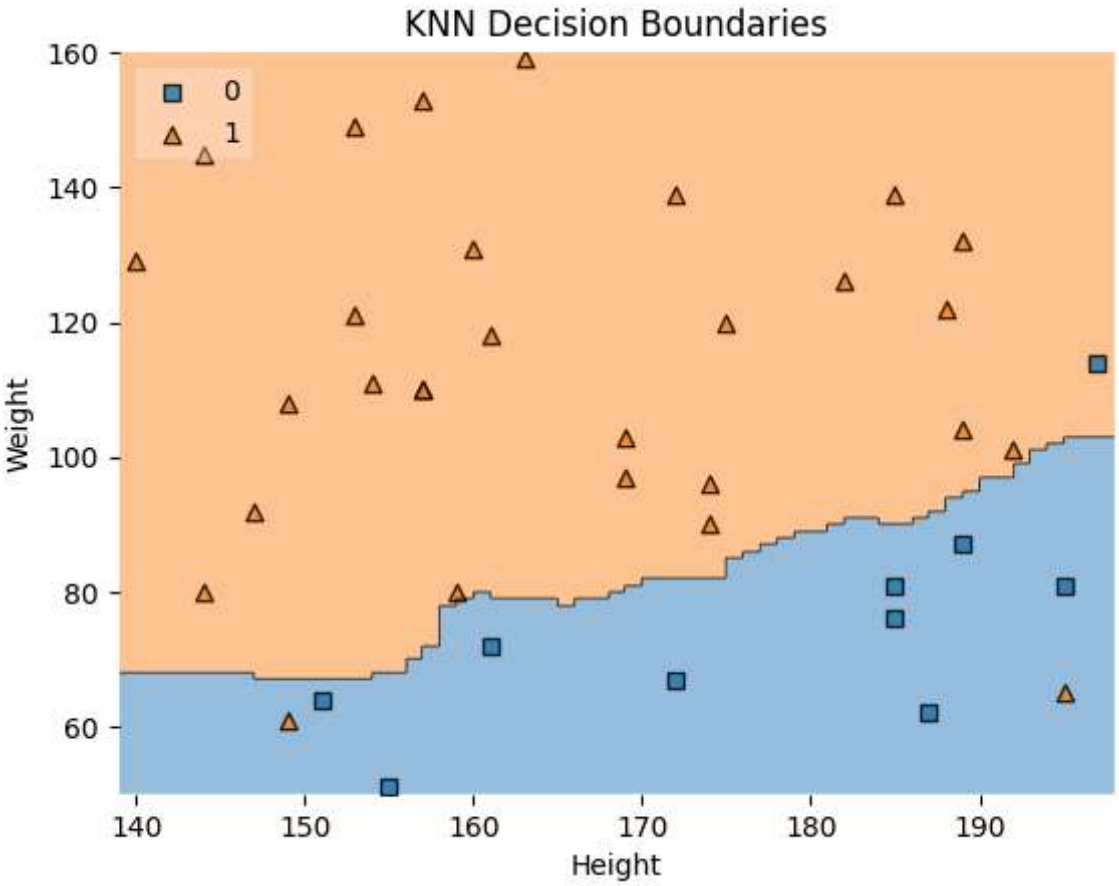
In [20]:

```
H = float(input("Enter Height: "))
W = float(input("Enter Weight: "))
predicted_label = knn.predict([[H, W]])
print("Predicted label:", predicted_label[0])
```

```
Predicted label: 0
```

In [21]:

```
knn.fit(X_train, y_train)
plot_decision_regions(X_train.values, y_train.values, clf=knn, legend=2)
plt.xlabel('Height')
plt.ylabel('Weight')
plt.title('KNN Decision Boundaries')
plt.show()
```



Result

The data set thus classified using k-Nearest Neighbour algorithm and printed correct and wrong predictions